Program 11.11

```
using namespace std;
#include<iostream>
#include<ctype.h>
#include<string.h>
int main()
{
        char str1[50];
        cout<<"\nType any string in Upper case :";
        gets(str1);
        cout<<"\n Converted the Source string “<<str1<<into Lower Case is "<<strlwr(str1);
}
```

Output:

```
        Type any string in Upper case : COMPUTER SCIENCE
Converted the Source string COMPUTER SCIENCE into lower Case is computer science
```

11.4.4 Mathematical functions (math.h)

Most of the mathematical functions are defined in math.h header file which includes basic mathematical functions.

11.4.4.1 cos() function

The cos() function takes a single argument in radians. The cos() function returns the value in the range of [-1, 1]. The returned value is either in double, float, or long double.

Program 11.12

```
#include <iostream>
#include <math.h>
using namespace std;
int main()
{
        double x = 0.5, result;
        result = cos(x);
        cout << "COS("<<x<<")= "<<result;
}
```

Output:

```
        COS(0.5)= 0.877583
```

11.4.4.2 sqrt() function

The **sqrt()** function returns the square root of the given value. The **sqrt()** function takes a single non-negative argument. If a **negative value** is passed as an argument to **sqrt()** function, a **domain error occurs**.

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